



Alongside AWS Professional Services, Casio Computer Developed a Modern Online Learning App in 4 Months, Shifting to Agile Development

Learn how manufacturing company Casio Computer reduced time to development and facilitated recurring customer engagement using a suite of AWS services.

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4 months

to develop educational web app

104 feature updates

in 15 months



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Overview

[Casio Computer](#) has introduced highly unique products to the world, such as electronic dictionaries and electronic musical instruments, as well as wristwatches like G-SHOCK. The company adopted Amazon Web Services (AWS) as a solution when releasing the online learning app ClassPad.net in the education sector. Engineers with zero experience building modern apps on AWS developed and released the service in only 4 months. The shift to agile development has changed their engineering styles and accelerated development speed.



Opportunity | Pushing Learning App Release 3 Years ahead of Schedule

For the New CASIO C30 Project, which aims to maximize corporate value in fiscal year 2030, Casio Computer is shifting to a recurring business model across three main business areas: watches, education, and musical instruments. “Instead of just selling a product, as we had once done, our purpose is to focus more on providing new values and experiences by engaging with customers after they purchase our products. To do so, we are expanding the scale and area of our business,” says Kazunori Yanagi, general manager of the development management department 2, technology division at Casio Computer. The company released the online learning app ClassPad.net in 2021 to create value for customer experiences through recurring engagement.

ClassPad.net is a cloud-based app developed based on the concept of “digitizing student desks” and makes use of the know-how of electronic dictionaries and scientific calculators, which are flagship products of the company. ClassPad.net was initially planned to be released in April 2024, in line with the completion of the GIGA School Initiative run by the Ministry of Education, Culture, Sports, Science, and Technology of Japan. However, when the implementation of the GIGA School Initiative was moved up due to the COVID-19 pandemic, the company also had to move up the development of ClassPad.net by 3 years.

As it was, the web application development department faced issues with platforms, architecture, development languages, as well as slower business speed due to waterfall-style development and development systems. Masaharu Meki, director of the development department 2, development management



department 2, technology division at Casio Computer says, "At that time, in 2020, the development department was divided by service, so cooperation between groups was difficult, and each was working independently. Therefore, we switched to a matrix-type organization that is responsible for multiple services, modernizing apps, and promoting agile development."

Solution | Building a Modern Architecture Using the Serverless Model and CI/CD

When the development department began developing ClassPad.net, it decided to adopt AWS to make the most of managed services. "The entire web app development team decided to use AWS as the foundation and consolidate its technology and know-how," says Meki. The team decided to adopt AWS based on a comprehensive assessment of industry share and performance, the abundance of publicly available technical information, the number of people with in-house development experience, and Casio Computer's enterprise-wide cloud policy.

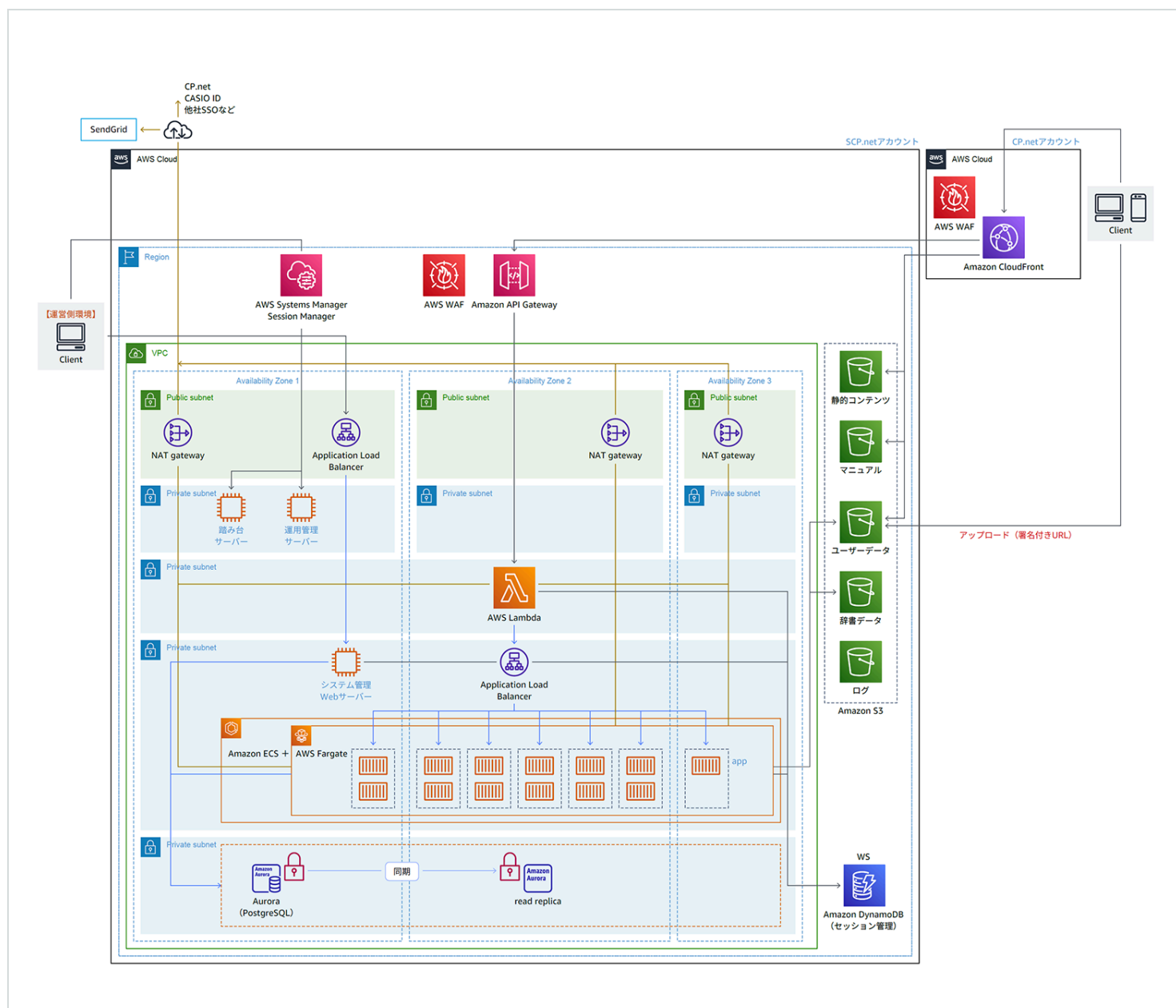
In this project, the company aimed for an optimal solution without compromising on the architecture and decided to improve development efficiency after release. However, when it began building in November 2020, Casio Computer didn't have experience developing modern apps on AWS. So, it decided to engage [AWS Professional Services](#) to get technical assistance. "To achieve our ideal architecture in a short period and to improve business speed, we adopted AWS Professional Services because, by doing so, our knowledge remained in-house," says Meki.

With the completion of the design phase in about 1 month with the support of AWS Professional Services, the team developed the front end, backend, and infrastructure in parallel. After testing in a production environment, Casio Computer released a beta version with some features in April 2021. The beta was built using [Amazon Elastic Container Service](#) (Amazon ECS) on [AWS Fargate](#). These services adopt [AWS Lambda](#) and [Amazon API Gateway](#) for Backends For Frontends (BFF) patterns and containers for backend microservices.

Additionally, the company is building a DevOps environment using the AWS Code series. Akihiko Masuda, manager of development 23, development department 2, development management department 2, technical division at Casio Computer says, "In parallel with development, we created guidelines and accumulated knowledge within the team. Our contact from AWS Professional Services consulted with us closely, and we were able to build the architecture without compromise."



Architecture



Outcome | Achieving 16 Releases in 15 Months due to Improved Development Efficiency

Since the beta launch in April 2021, ClassPad.net has already been introduced to over 600 high schools. Casio Computer continued to add features after release, and in the 15 months until September 2022, it made 16 releases, implemented 104 feature upgrades, and closed 4,672 tickets. These benefits were made possible through automated infrastructure construction through infrastructure as code (IaC) and automated deployment through continuous integration and continuous deployment (CI/CD). "IaC has greatly reduced the burden of building an environment and made it easier to repurpose and reproduce infrastructure. With CI/CD, we shortened deployment time from 1 day to around 1 hour, and we have had virtually no deployment issues," says Masuda.

Furthermore, serverless and managed services facilitated a reduction in resource management, and containerization freed teams from scalability constraints in the event of increased app users. The engineers were also more motivated by developing a modern app, bringing about a change in mindset throughout the development department. "The biggest result of this project is that we built a system where engineers can acquire new skills and continuously develop better services in addition to recurring business," says Yanagi.



Shifting to function-based scrum development, the development department released human resources management system Hito-Compass in June 2022 by reapplying know-how acquired through the development of ClassPad.net and existing assets. It is now developing other commercial web applications as well. In addition to continuing to add features to ClassPad.net, the company began overseas expansion in November 2022 and provides features for overseas markets.

Casio Computer planned to promote cloud transformation centered around AWS services as a company-wide integrated digital transformation solution. It began the project in earnest in 2022. Katsuhiko Abukawa, director of the integrated platform department of the digital control department at Casio Computer says, “To actualize the customer-centered value chain set forth in the management policy and continue to provide new value, it is necessary to have an environment where business divisions can focus on customer needs. To that end, to achieve rapid development and reduced operational load, we plan to adopt AWS as the common solution for our infrastructure and for the digital control department to collectively provide various functions such as development architects and observability. With the cooperation from various departments, we’ll establish an organization that promotes cloud usage, such as cloud center of excellence, by fiscal year 2023. We want to continue to modernize new businesses and services.”

About Casio Computer

In 1957, Casio Computer was established and developed the world’s first compact all-electric calculator, CASIO 14-A. Since then, the company has provided products with unprecedented value based on the corporate creed of “creativity and contribution.” The business consists of watches—the leading pillar under the brand G-SHOCK that accounts for 60 percent of its global sales—educational products, musical instruments, systems, and five other business pillars. Casio Computer also runs business overseas, accounting for 75 percent of its sales.



Kazunori Yanagi



Meki Masaharu



Akihiko Masuda



Katsuhiko Abugawa

AWS Services Used

AWS Professional Services

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Amazon Elastic Container Service

Amazon Elastic Container Service (ECS) is a fully managed container orchestration service that helps you to more efficiently deploy, manage, and scale containerized applications.

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AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

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AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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